AMA515A Assignment 1

1. The error in forecasting for period T+1 at the end of period T is defined as e(T+1) = $x_{T+1} - \hat{x}_{T+1}(T)$. If the observations are independent random variables, x_{T+1} will be independent of $\hat{x}_{T+1}(T)$. Assume simple (single) exponential smoothing is applied to the constant process $x_t = b + \varepsilon_t$, where var(ε_t) = σ_ε^2. Show that the variance of the forecast error, σ_e^2, satisfies the relationship $\sigma_e^2 = \frac{2}{2-\alpha}\sigma_\varepsilon^2$.

2. (a). Two processes { Z_t } and { Y_t } are said to be independent if for any time points $t_1, \ldots, t_m$ and $s_1, \ldots, s_n$ the random variables $\{Z_{t_1}, \ldots, Z_{t_m}\}$ are independent of the variables $\{Y_{s_1}, \ldots, Y_{s_n}\}$. Show that if $\{Z_t\}$ and $\{Y_t\}$ are independent stationary processes, then $W_t = Z_t + Y_t$ is stationary.

   (b). Let $\{X_t\}$ be a time series. However, because the measurement process itself is not perfect, we actually observe $Y_t = X_t + e_t$. Assume that X_t and e_t are independent processes. We call X_t the signal and e_t the measurement noise. If $\{X_t\}$ is stationary with autocorrelation function ρ_k, show that $\{Y_t\}$ is also stationary with

   $$Corr(Y_t, Y_{t-k}) = \frac{\rho_k}{1 + \sigma^2_e / \sigma^2_X} \quad \text{for } k \geq 1.$$

   We call σ^2_X / σ^2_e the signal-to-noise ratio (SNR). Note that the larger the SNR, the closer the autocorrelation function of the observed process $\{Y_t\}$ is to the autocorrelation function of the desired signal $\{X_t\}$.

3. The electricity consumption of a certain city (in coded units) from 1991 to 1997 is given in the following table.

Year	Time index (t)	data
91	1	676
92	2	825
93	3	774
94	4	716
95	5	940
96	6	1159
97	7	1384

   (a) It is felt that the data can be fitted by the following linear model:
   $X_t = a + b_1 t + e_t$, where e_t is NID(0,σ^2).

The Durbin Watson statistic is defined by d= $\sum_{t=2}^{n} (\hat{e}_t - \hat{e}_{t-1})^2 / \sum_{t=1}^{n} \hat{e}_t^2$, where $\hat{e}_t$ is the residual after fitting the model to the data. It's used to test whether the residuals are autocorrelated or not. The residuals for the data above are:

t	Residuals $\hat{e}_t$
1	68.071
2	111.429
3	-45.214
4	-208.857
5	-90.5
6	22.857
7	142.214

Using the given residuals, evaluate the Durbin Watson statistic for this model. What is your conclusion from the value of the test statistic?

(b) Let $\{y_T\}$ be a time series. The Holt's double exponential smoothing algorithm for a trend model is defined as follows:

1. Obtain an updated estimate $a_0(T)$ of the intercept component by using the equation
$a_0(T) = \alpha y_T + (1-\alpha)[a_0(T-1) + b_1(T-1)]$, where $0<\alpha<1$

2. Obtain an updated estimate $b_1(T)$ of the slope component by using the equation

$b_1(T) = \beta[a_0(T) - a_0(T-1)] + (1-\beta)b_1(T-1)$, where $0<\beta<1$

3. A one-step ahead forecast of the future value y_{T+1} made at time T is

$\hat{y}_T(1) = a_0(T) + b_1(T)$

For the above data, it is given $a_0(0) = 502.286, b_1(0) = 105.643$. $\alpha = 0.2; \beta = 0.1$.

(i) By using Holt's algorithm, find the one-step-ahead forecasts for the period 1992 to 1996, and calculate the MAE (mean absolute error) of the forecasts.
(ii) Find the naive forecasts in the same period and calculate the MAE. By comparing the MAE of these two methods, determine which one is better.

4. The following data set is the amount of coffee (in coded units) produced by a certain company in 2006.

month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
data	200	145	195	197.5	210	175	155	140

A forecaster thinks that a constant model, $X_t = b_1 + a_t$, where $a_t \sim NID(0, \sigma^2)$, is appropriate for the data. He then suggests to use the method of Simple Exponential Smoothing, namely $S_t = \alpha X_t + (1-\alpha)S_{t-1}$, to obtain forecasts for this set of data. Here $0 < \alpha < 1$.

(a) Using the January observation as S_0, calculate the Simple Exponential Smoothing forecasts for Feb to Aug. Let $\alpha = 0.4$.

(b) It is given that $\text{var}(X_{t+1} - S_t) = \left(\frac{2}{2-\alpha}\right)\sigma^2$. Based on your result in (a), find an estimate of σ^2.

(c) Find an estimate of $\text{var}(S_t)$.

(d) Use the Simple Exponential Smoothing algorithm to forecast the coffee production in September 2006, and find a 95% confidence interval for the forecast. You may assume the forecast follows an approximate normal distribution.

5. Let $\{Z_t\}$ be a sequence of independent normal random variables, each with mean 0 and variance σ^2 and let a, b and c be constants. Which, if any, of the following processes are stationary? For each stationary process specify the mean and autocovariance function (which includes the variance).

a. $X_t = a + bZ_t + cZ_{t-2}$
b. $X_t = Z_1 \cos(ct) + Z_2 \sin(ct)$
c. $X_t = Z_t \cos(ct) + Z_{t-1} \sin(ct)$
d. $X_t = a + bZ_0$
e. $X_t = Z_0 \cos(ct)$

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